Operating System (ISE3137)

Jump Together, Fly Farther!

School of Global Convergence Studies

인하대학교 국제학부

Week 3 Lab

ISE Department

Prof. Mehdi Pirahandeh

LINUX FILE SYSTEM HIERARCHY

THE FILE-SYSTEM HIERARCHY

All files on a Linux system are stored on file systems, which are organized into a single *inverted* tree of directories, known as a *file-system hierarchy*. This tree is inverted because the root of the tree is said to be at the *top* of the hierarchy, and the branches of directories and subdirectories stretch *below* the root.

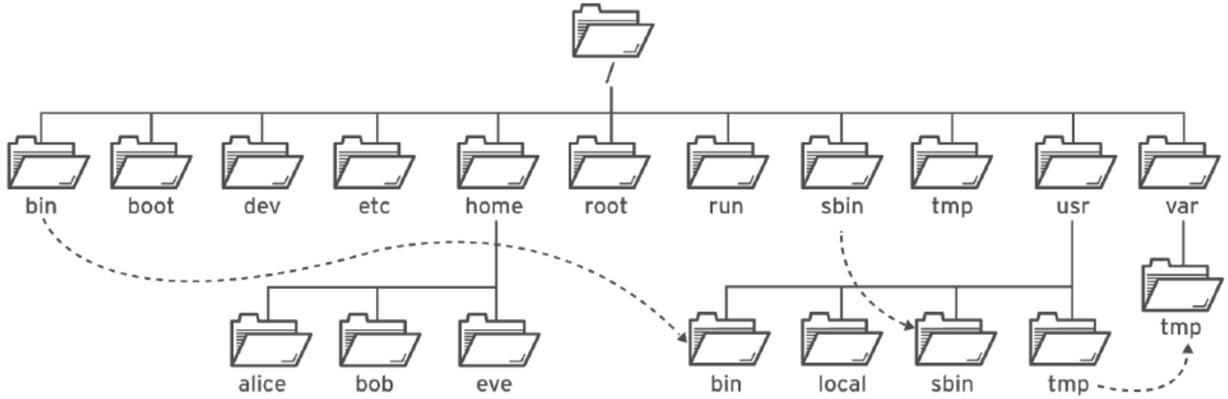


Figure 2.1: Significant file-system directories in Red Hat Enterprise Linux 8

The / directory is the root directory at the top of the file-system hierarchy. The / character is also used as a *directory separator* in file names. For example, if **`etc`** is a subdirectory of the / directory, you could refer to that directory as **`/etc`**. Likewise, if the **`/etc`** directory contained a file named **`issue`**, you could refer to that file as **`/etc/issue`**.

Subdirectories of / are used for standardized purposes to organize files by type and purpose. This makes it easier to find files. For example, in the root directory, the subdirectory **`/boot`** is used for storing files needed to boot the system.

LINUX FILE SYSTEM HIERARCHY

NOTE

The following terms help to describe file-system directory contents:

- *static* content remains unchanged until explicitly edited or reconfigured.
- *dynamic* or *variable* content may be modified or appended by active processes.
- *persistent* content remains after a reboot, like configuration settings.
- *runtime* content is process- or system-specific content that is deleted by a reboot.

LINUX FILE SYSTEM HIERARCHY

인하대학교 INHA UNIVERSITY

The following table lists some of the mo st important directo ries on the system b y name and purpose .

LOCATION	PURPOSE
/usr	Installed software, shared libraries, include files, and read-only program data. Important subdirectories include: • **/usr/bin**: User commands. • **/usr/sbin**: System administration commands. • **/usr/local**: Locally customized software.
/etc	Configuration files specific to this system.
/var	Variable data specific to this system that should persist between boots. Files that dynamically change, such as databases, cache directories, log files, printer-spooled documents, and website content may be found under **/var**.
/run	Runtime data for processes started since the last boot. This includes process ID files and lock files, among other things. The contents of this directory are recreated on reboot. This directory consolidates **/var/run** and **/var/lock** from earlier versions of Red Hat Enterprise Linux.
/home	*Home directories* are where regular users store their personal data and configuration files.
/root	Home directory for the administrative superuser, `root`.
/tmp	A world-writable space for temporary files. Files which have not been accessed, changed, or modified for 10 days are deleted from this directory automatically. Another temporary directory exists, **/var/tmp**, in which files that have not been accessed, changed, or modified in more than 30 days are deleted automatically.
/boot	Files needed in order to start the boot process.
/dev	Contains special *device files* that are used by the system to access hardware.

FILE MAGEMENT

ACTIVITY	COMMAND SYNTAX
Create a directory	`mkdir` *`directory`*
Copy a file	`cp` *`file new-file`*
Copy a directory and its contents	`cp -r` *`directory new-directory`*
Move or rename a file or directory	`mv` *`file new-file`*
Remove a file	`rm` *`file`*
Remove a directory containing files	`rm -r` *`directory`*
Remove an empty directory	`rmdir` *`directory`*

FILE MAGEMENT

인하대학교
INHA UNIVERSITY

```
[user@host ~]$ mkdir Videos/Watched
[user@host ~]$ ls -R Videos
Videos/:
blockbuster1.ogg  blockbuster2.ogg  Watched

Videos/Watched:
```

In the following example, files and directories are organized beneath the **/home/user/Documents** directory. Use the **mkdir** command and a space-delimited list of the directory names to create multiple directories.

```
[user@host ~]$ cd Documents
[user@host Documents]$ mkdir ProjectX ProjectY
[user@host Documents]$ ls
ProjectX  ProjectY
```

Use the **mkdir -p** command and space-delimited relative paths for each of the subdirectory names to create multiple parent directories with subdirectories.

```
[user@host Documents]$ mkdir -p Thesis/Chapter1 Thesis/Chapter2 Thesis/Chapter3
[user@host Documents]$ cd
[user@host ~]$ ls -R Videos Documents
Documents:
ProjectX  ProjectY  Thesis

Documents/ProjectX:

Documents/ProjectY:
```

Copying Files

The **cp** command copies a file, creating a new file either in the current directory or in a specified directory. It can also copy multiple files to a directory.

WARNING

If the destination file already exists, the **cp** command overwrites the file.

```
[user@host ~]$ cd Videos
[user@host Videos]$ cp blockbuster1.ogg blockbuster3.ogg
[user@host Videos]$ ls -l
total 0
-rw-rw-r--. 1 user user    0 Feb  8 16:23 blockbuster1.ogg
-rw-rw-r--. 1 user user    0 Feb  8 16:24 blockbuster2.ogg
-rw-rw-r--. 1 user user    0 Feb  8 16:34 blockbuster3.ogg
drwxrwxr-x. 2 user user 4096 Feb  8 16:05 Watched
[user@host Videos]$
```

When copying multiple files with one command, the last argument must be a directory. Copied files retain their original names in the new directory. If a file with the same name exists in the target directory, the existing file is overwritten. By default, the **cp** does not copy directories; it ignores them.

FILE MAGEMENT

In the following example, two directories are listed, **Thesis** and **ProjectX**. Only the last argument, **ProjectX** is valid as a destination. The **Thesis** directory is ignored.

```
[user@host Videos]$ cd ../Documents
[user@host Documents]$ cp thesis_chapter1.odf thesis_chapter2.odf Thesis ProjectX
cp: omitting directory `Thesis'
[user@host Documents]$ ls Thesis ProjectX
ProjectX:
thesis_chapter1.odf  thesis_chapter2.odf

Thesis:
Chapter1  Chapter2  Chapter3
```

In the first **cp** command, the **Thesis** directory failed to copy, but the **thesis_chapter1.odf** and **thesis_chapter2.odf** files succeeded.

If you want to copy a file to the current working directory, you can use the special **.** directory:

```
[user@host ~]$ cp /etc/hostname .
[user@host ~]$ cat hostname
host.example.com
[user@host ~]$
```

Use the copy command with the **-r** (*recursive*) option, to copy the **Thesis** directory and its contents to the **ProjectX** directory.

FILE MAGEMENT

In the following example, two directories are listed, **Thesis** and **ProjectX**. Only the last argument, **ProjectX** is valid as a destination. The **Thesis** directory is ignored.

```
[user@host Videos]$ cd ../Documents
[user@host Documents]$ cp thesis_chapter1.odf thesis_chapter2.odf Thesis ProjectX
cp: omitting directory `Thesis'
[user@host Documents]$ ls Thesis ProjectX
ProjectX:
thesis_chapter1.odf  thesis_chapter2.odf

Thesis:
Chapter1  Chapter2  Chapter3
```

In the first **cp** command, the **Thesis** directory failed to copy, but the **thesis_chapter1.odf** and **thesis_chapter2.odf** files succeeded.

If you want to copy a file to the current working directory, you can use the special **.** directory:

```
[user@host ~]$ cp /etc/hostname .
[user@host ~]$ cat hostname
host.example.com
[user@host ~]$
```

Use the copy command with the **-r** (*recursive*) option, to copy the **Thesis** directory and its contents to the **ProjectX** directory.

FILE MAGEMENT

```
[user@host Documents]$ cp -r Thesis ProjectX
[user@host Documents]$ ls -R ProjectX
ProjectX:
Thesis  thesis_chapter1.odf  thesis_chapter2.odf

ProjectX/Thesis:
Chapter1  Chapter2  Chapter3

ProjectX/Thesis/Chapter1:

ProjectX/Thesis/Chapter2:
thesis_chapter2.odf

ProjectX/Thesis/Chapter3:
```

FILE MAGEMENT

Moving Files

The **mv** command moves files from one location to another. If you think of the absolute path to a file as its full name, moving a file is effectively the same as renaming a file. File contents remain unchanged.

Use the **mv** command to *rename* a file.

```
[user@host Videos]$ cd ../Documents
[user@host Documents]$ ls -l thesis*
-rw-rw-r--. 1 user user 0 Feb  6 21:16 thesis_chapter1.odf
-rw-rw-r--. 1 user user 0 Feb  6 21:16 thesis_chapter2.odf
[user@host Documents]$ mv thesis_chapter2.odf thesis_chapter2_reviewed.odf
[user@host Documents]$ ls -l thesis*
-rw-rw-r--. 1 user user 0 Feb  6 21:16 thesis_chapter1.odf
-rw-rw-r--. 1 user user 0 Feb  6 21:16 thesis_chapter2_reviewed.odf
```

Use the **mv** command to *move* a file to a different directory.

```
[user@host Documents]$ ls Thesis/Chapter1
[user@host Documents]$
[user@host Documents]$ mv thesis_chapter1.odf Thesis/Chapter1
[user@host Documents]$ ls Thesis/Chapter1
thesis_chapter1.odf
[user@host Documents]$ ls -l thesis*
-rw-rw-r--. 1 user user 0 Feb  6 21:16 thesis_chapter2_reviewed.odf
```

Removing Files and Directories

The `rm` command removes files. By default, `rm` will not remove directories that contain files, unless you add the `-r` or `--recursive` option.

IMPORTANT

There is no command-line undelete feature, nor a "trash bin" from which you can restore files staged for deletion.

FILE MAGEMENT

It is a good idea to verify your current working directory before removing a file or directory.

```
[user@host Documents]$ pwd
/home/student/Documents
```

Use the **rm** command to remove a single file from your working directory.

```
[user@host Documents]$ ls -l thesis*
-rw-rw-r--. 1 user user 0 Feb  6 21:16 thesis_chapter2_reviewed.odf
[user@host Documents]$ rm thesis_chapter2_reviewed.odf
[user@host Documents]$ ls -l thesis*
ls: cannot access 'thesis*': No such file or directory
```

If you attempt to use the **rm** command to remove a directory without using the **-r** option, the command will fail.

```
[user@host Documents]$ rm Thesis/Chapter1
rm: cannot remove `Thesis/Chapter1': Is a directory
```

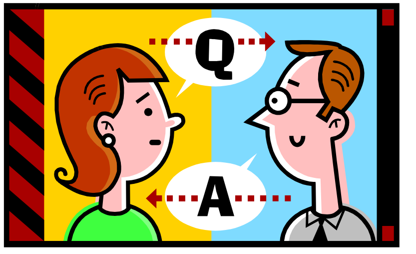
Use the **rm -r** command to remove a subdirectory and its contents.

```
[user@host Documents]$ ls -R Thesis
Thesis/:
Chapter1  Chapter2  Chapter3

Thesis/Chapter1:
```

Brief break (if on schedule)

Q&A?



Prof. Mehdi Pirahandeh
E-mail : mehdi at inha dot ac dot kr